MAT237 Notes

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'25 Fall - '26 Winter

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§1 Day 1: Administrative Stuff (Sept 2, 2025)

Everything is in the syllabus, but we did play with some blocks! There's many different ways to visualize the same thing. Went over classroom norms and whatnot, and then looked at syllabus, no math content today.

About these notes: I keep track of things mentioned in lecture. You can likely find more detail in the textbook.

§2 Day 2: Speed and Velocity (Sept 4, 2025)

We want to differentiate and integrate functions $A \to B$, where $A \subseteq \mathbb{R}^m$, $B \subseteq \mathbb{R}^n$. Today we study the case where $m = 1$, so functions $A \subseteq \mathbb{R} \to \mathbb{R}^n$. Today, will mainly look at functions $\mathbb{R} \to \mathbb{R}^n$, as having a single parameter makes them much easier to work with.

Recall that distance is a scalar quantity, while velocity is a vector, meaning it has both magnitude and direction. Have $f : \mathbb{R} \to \mathbb{R}^n$ model some particle's position, and $||\cdot||$ be the euclidean norm. The average speed[1] over the time interval t_1 and t_2 ($t_1 < t_2$) is given by

$$\frac{||f(t_2) - f(t_1)||}{t_2 - t_1}$$

The instantaneous speed at time t is given by

$$\lim_{h \to 0} \left|\left| \frac{f(t+h) - f(t)}{h} \right|\right|$$

The average velocity between t_1 and t_2 is given by

$$\frac{f(t_2) - f(t_1)}{t_2 - t_1}$$

and the instantaneous velocity at t by

$$\lim_{h \to 0} \frac{f(t+h) - f(t)}{h}$$

Theorem 2.1 (Absolute Homogeneity of Euclidean Norm)

The euclidean norm has the *absolute homogeneity* property.

For all $v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{R}^n$, a scalar λ

$$||\lambda v|| = |\lambda| \, ||v||$$

2.1 and various properties of norms were not mentioned in class.[2]

Proof.

$$\begin{aligned} ||\lambda v|| &= \left|\left| \begin{pmatrix} \lambda v_1 \\ \vdots \\ \lambda v_n \end{pmatrix} \right|\right| \\ &= \sqrt{\lambda^2 {v_1}^2 + \cdots + \lambda^2 {v_n}^2} \\ &= \sqrt{\lambda^2 ({v_1}^2 + \cdots + {v_n}^2)} \\ &= |\lambda| \sqrt{{v_1}^2 + \cdots + {v_n}^2)} \\ &= |\lambda| \, ||v|| \end{aligned}$$

□

[1] I don't believe we are concerned with the 'actual' speed over time in this course (as in the physics definition), since that would involve finding the arc length and even more trouble. Just assume it means 'magnitude of the displacement vector'.

[2] I included this for completeness because people didn't believe that $\frac{||\gamma(6+h) - \gamma(6)||}{|h|}$ and $\left|\left| \frac{\gamma(6+h) - \gamma(6)}{h} \right|\right|$ were the same quantity.

§3 Day 3: Graphs, Level sets, and Slices (Sept 8, 2025)

Office hours starting next week! Check Quercus for more details.

Last class, we looked at functions $\mathbb{R} \to \mathbb{R}^n$. Today we look at functions $\mathbb{R}^n \to \mathbb{R}$. We develop graphs, level sets, and slices because they offer new ways to analyze and study properties of such functions, that cannot be easily captured otherwise. For example, it is difficult to visualize a more than 3 dimensional vector.
Have $A \subseteq \mathbb{R}^n$, $f : A \to \mathbb{R}^n$.

Definition 3.1 (Graph)**.** The graph of f is $\{(x, f(x)) : x \in A\}$

Note that when you plot a function $f : A \subseteq \mathbb{R}^n \to \mathbb{R}$, the graph would exist in $\mathbb{R}^{n+1}$.

Definition 3.2 (Level Set)**.** The level set of f at k is $\{x \in \mathbb{R}^n : f(x) = k\}$[3]

To produce the **slice** of a graph, we need to hold a coordinate x_i constant, which we set to a. We call the following a x_i-slice, where the c is at the i-th position in the tuple:

$$\{(x_1, \cdots, x_{n+1}) \in \mathbb{R}^n : (x_1, \cdots, a, \cdots, x_n) \in A, x_{n+1} = f(x_1, \cdots, a, \cdots, x_n)\}$$

where in the first tuple, x_i is omitted, with x_i is replaced by a in the second and third tuples. Note that the slice lives in $\mathbb{R}^n$, because we already have the information that $x_i = a$.

In this course, you always want to specify the domain and codomain of your function, to avoid confusion.

Problem 3.3. Give a function $f : \mathbb{R}^2 \to \mathbb{R}$ whose level set at 0 is the set

$$\{(x, y) \in \mathbb{R}^2 : |x| = |y|\}$$

To solve these kinds of problems, you set $0 = |y| - |x|$ and you would get a candidate function $f(x, y) = |y| - |x|$. To show that two sets A and B are equal, you would typically prove that $A \subseteq B$, and $B \subseteq A$.

Alternatively you could define a function

$$g(x, y) = \begin{cases} 0 & \text{if } |x| = |y| \\ 1 & \text{otherwise} \end{cases}$$

which satisfies the requirements by construction.

[3] If $A \subseteq \mathbb{R}^2$, the level set is called a *contour*.

§4 Day 4: Vector Fields and Transformations (Sept 9, 2025)

So far, we have seen parametric curves ($\mathbb{R} \to \mathbb{R}^n$), real valued functions ($\mathbb{R}^n \to \mathbb{R}$). Today we look at functions $\mathbb{R}^n \to \mathbb{R}^n$, which are called vector fields or transformations[4].

Definition 4.1 (Vector Field)**.** A n dimensional vector field is a function $F : A \to B$ with $A, B \subseteq \mathbb{R}^n$.

Note that vector fields are capitalized as per convention.

(Not testable) Newton's law of gravity states that the force exerted by an object at the origin with mass m_1 on an object at (x, y, z) with mass m_2 is given by

$$F(x, y, z) = \frac{-Gm_1m_2}{||(x, y, z)||^2} \cdot \frac{(x, y, z)}{||(x, y, z)||}$$

The magnitude is controlled by the first part of the product (note that there are only scalars), and the direction is controlled by the unit vector (note that it is scaled down to have a magnitude of 1).

[4] very uncommon to call such functions transformations

§5 Day 5: Coordinate Transformations (Sept 11, 2025)

Definition 5.1 (Coordinate Transformation)**.** A **coordinate transformation** $f : A \to B$ is a continuous transformation that is usually bijective. A and the map f form a **coordinate system** for the codomain B.

We want to plot subsets of B and describe them using the coordinate system defined by f and A. We use (u, v) to describe the elements in A, and (x, y) for B, and would write

$$g(u, v) = (u^2 + v^2, v)$$

or simply

$$(x, y) = (u^2 + v^2, v)$$

Definition 5.2 (Polar Coordinate Transformation)**.** A map $T : \mathbb{R}^2 \to \mathbb{R}^2$, with

$$T(r, \theta) = (r \cos\theta, r \sin\theta)$$

T describes a map from polar coordinates to cartesian coordinates. The radius can be negative. Two notable properties are $T(r, \theta) = T(-r, \theta + \pi)$ and $T(r, \theta) = T(r, \theta + 2\pi)$, following from trigonometry.

The set $\{(r, \theta) \in \mathbb{R}^2 : r = 2\}$ would describe the set $\{(2\cos\theta, 2\sin\theta) \in \mathbb{R}^2 : \theta \in \mathbb{R}\}$, which corresponds to a circle of radius 2 centered at the origin. Restricting both sets to $\theta \in \mathbb{R}^+$ or $\theta \in \mathbb{R}^-$ would still correspond to the same set, meaning that the polar coordinate transformation is not injective. Another way to show is to consider the case $r = 0$.

What follows was not covered in lecture:

Definition 5.3 (Cylindrical Coordinate Transformation)**.** A map $T : \mathbb{R}^3 \to \mathbb{R}^3$, with

$$T(r, \theta, z) = (r \cos\theta, r \sin\theta, z)$$

This is similar to the polar coordinate transformation except we add an additional z field which remains unchanged.

Definition 5.4 (Spherical Coordinate Transformation)**.** A map $T : \mathbb{R}^3 \to \mathbb{R}^3$ with

$$T(\rho, \theta, \phi) = (\rho \cos\theta \sin\phi, \rho \sin\theta \sin\phi, \rho \cos\phi)$$

If this gets covered in class I'll make a writeup deriving the formula.

§6 Day 6: Parametric, Implicit, and Explicit Form (Sep 15, 2025)

Have $n < m$ be positive integers.

Definition 6.1 (Parametric Form)**.** A set $S \subseteq \mathbb{R}^m$ can be written in parametric form (with n-variables) if there exists a set $A \subseteq \mathbb{R}^n$, and a continuous map $g : A \to \mathbb{R}^m$ such that

$$S = \{g(x) : x \in A\}$$

Definition 6.2 (Higher Dimensional Graphs)**.** Let $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^m$ be a continuous function.

$$\text{graph}(f) = \{(x, f(x)) \in \mathbb{R}^n \times \mathbb{R}^m \mid x \in A\}$$

In this course $\mathbb{R}^a \times \mathbb{R}^b = \mathbb{R}^{a+b}$, and we don't care whether the nesting of ordered pairs is done in the first or second entry.

Definition 6.3 (Explicit Form)**.** A set S can be written in explicit form in n variables if $S \subseteq \mathbb{R}^m$ is $\text{graph}(f)$ for some continuous function $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^m$.

Definition 6.4 (Implicit Form)**.** A set S can be written in implicit form in n variables if there exists a constant $c \in \mathbb{R}^m$, and a continuous function $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^m$ such that $S = f^{-1}(\{c\})$.

Implicit form is a generalized version of a level set. The level set is defined for functions $g : \mathbb{R}^n \to \mathbb{R}$, notice that $g^{-1}(\{c\}) = \{x \in \mathbb{R}^n : g(x) = c\}$ is the same as the level set of g at c.

§7 Day 7: Interior, Boundary, and Closure (Sep 16, 2025)

Let $A \subseteq \mathbb{R}^n$.

Definition 7.1 (Interior Point)**.** $p \in \mathbb{R}^n$ is an interior point f A if there exists an $\epsilon > 0$ such that $B_\epsilon(p) \subseteq A$.

Definition 7.2 (Interior)**.** The interior denoted A^o is the set of all interior points of A.

Definition 7.3 (Boundary Point)**.** $p \in \mathbb{R}^n$ is a boundary point if for every $\epsilon > 0$, $B_\epsilon(p) \cap A$ and $B_\epsilon \cap A^c$ are non-empty.

The boundary point can be isolated, there may exist ϵ-balls at p such that p is the only point in said ball.

Definition 7.4 (Topological Boundary)**.** The topological boundary of A, written ∂A.

Definition 7.5 (Limit Point)**.** $p \in \mathbb{R}^n$ is a limit point of A if for every $\epsilon > 0$, $B_\epsilon \setminus \{p\}$ contains points in A. The set of all limit points is written A^*.

A set B containing points in A is the same as saying that $A \cap B \neq \emptyset$. Also to trip you up, sometimes questions will need you negate the definition, so do that if it seems unintuitive to explain.

Definition 7.6 (Closure)**.** The closure of A is written $\overline{A}$, defined as $\overline{A} = A^* \cup A$.

Theorem 7.7. Every interior point of A is a limit point of A. That is $A^o \subseteq A^*$.

Proof. Let p be an interior point of A. By definition, exist $\epsilon > 0$, such that $B_\epsilon(p) \subseteq A$. Let $\epsilon' > 0$ be arbitrary. If $\epsilon' > \epsilon$, note that $B_\epsilon(p) \subseteq B_{\epsilon'}(p)$e can take any point in the open ball $B_\epsilon(p) \setminus \{p\}$, which is a subset of A following from the definition. If $\epsilon' \leq \epsilon$, we can take any point in $B_{\epsilon'}(p) \setminus \{p\} \subseteq B_\epsilon(p) \setminus \{p\} \subseteq A$. □

Note that this proof relies on the fact for $\epsilon > 0$, there exist points in $B_\epsilon(p) \setminus \{p\}$ that are not p. This is not true from some topologies.

Theorem 7.8.

$$A^o \subseteq A \subseteq \overline{A}$$
$$A^o \cap \partial A = \emptyset$$
$$\overline{A} = A^o \cup \partial A$$
$$\partial A = \overline{A} \setminus A^o$$

You should be able to verify this by wrangling definitions.

§8 Day 8: Sequences (Sep 18, 2025)

Definition 8.1 (Sequence)**.** A sequence in in $\mathbb{R}^n$ is a function with domain $\{k \in \mathbb{Z} : k > k_0\}$ for some fixed $k_0 \in \mathbb{Z}$ and codomain $\mathbb{R}^n$.

Definition 8.2 (Subsequence)**.** Let $x : \mathbb{N}^+ \to \mathbb{R}^n$ be a sequence and $m : \mathbb{N}^+ \to \mathbb{N}^+$ be a strictly increasing function. The sequences $\{x(m(k))\}_{k=1}^{\infty}$ is a subsequence of the sequences $\{x(k)\}_{k=1}^{\infty}$.

Definition 8.3 (Convergence)**.** We fix a $p \in \mathbb{R}^n$. A sequence $\{x(k)\}_k$ in $\mathbb{R}^n$ converges to p if for every $\epsilon \in \mathbb{R}^+$, there exists a $K \in \mathbb{N}$ such that for every $k \in \mathbb{N}$, if $k \geq K$ then $||x(k) - p|| < \epsilon$.

Theorem 8.4. A sequence $\{x(k)\}$ converges to a point p if and only if for every $\epsilon > 0$, the set of indices $\{k \in \mathbb{N}^+ : x(k) \notin B_\epsilon(p)\}$ is finite.

Proof. ($\Longrightarrow$) Suppose $\{x(k)\}$ converges to p. Then we take an arbitrary $\epsilon > 0$. There exists a $K \in \mathbb{N}^+$ such that for $k \in \mathbb{N}^+$, $k \geq K$, $x(k)$ must belong to $B_\epsilon(p)$. Then points $x(j)$ not belonging to $B_\epsilon(p)$ must be of the form $x(j)$, where $j \in \mathbb{N}^+, j < k$. Since $j \in \{1, 2, \cdots, k-1\}$, we have finitely many such points.

($\Longleftarrow$) Suppose that for any $\epsilon > 0$ there exists only finitely many indices such that their corresponding point is not in $B_\epsilon(p)$. Let S denote the aforementioned set of indices. In the case where this is the empty set, we can take $K = 1$. If the set is non-empty, then take K as the largest such index plus one.

In both cases, all indices $k \geq K$ have $x(k) \in B_\epsilon(p)$. We can prove this by contradiction: Suppose there exists some $j \geq K$, $x(j) \notin B_\epsilon(p)$. If $S = \emptyset$, then this is immediately a contradiction. Otherwise, then the existence of j contradicts the maximality of K, since K is defined as the largest such index plus one.

As such a j does not exist, we have that for all $k \in \mathbb{N}^+$, if $k \geq K$ then $x(k) \in B_\epsilon(p)$. $\square$

We cannot say that $\{x(k)\}$ converges to p is equivalent to for all $\epsilon > 0$, the set $\{x(k) : k \in \mathbb{N}^+, x(k) \notin B_\epsilon(p)\}$ is finite. Consider the sequence $k \in \mathbb{N}^+$, $x(k) = (-1)^k$. This sequence has only finitely many points in its image, being $\{1, -1\}$, but is clearly not convergent to any point $p \in \mathbb{R}$ (picking $\epsilon < \frac{1}{2}$ would work).

§9 Day 9: Open and Closed Sets (Sep 22, 2025)

You should know that there exist sequential formulations equivalent to the definition of closed and open sets.

We want sets that have nice properties: Let $A \subseteq \mathbb{R}^n$. We want a class of sets such that if a sequence in $\mathbb{R}^n$ converges to $a \in A$, then the *tail* of the sequence belongs in A.

Definition 9.1 (Open Set)**.** A is open if every point of A is an interior point of A.

This is equivalent to saying $A = A^o$, and $A \cap \partial A = \emptyset$.

Definition 9.2 (Cloesd Set)**.** A is closed if every limit point of A belongs to A.

That is, $A^* \subseteq A$.

Theorem 9.3

A set A is open if and only if its complement $A^c = \mathbb{R}^n \setminus A$ is closed.

§10 Day 10: Compactness (Sep 23, 2025)

Definition 10.1 (Compactness)**.** A set $A \subseteq \mathbb{R}^n$ is compact if every sequence of A has a subsequence which converges to a point lying inside A.

This definition is actually for sequentially compact sets. For this reason, our theorem relating compactness to being closed and bounded will actually be the Bolzano-Weierstrass theorem instead of the usual Heine-Borel.[5]

Definition 10.2 (Boundedness)**.** $A \subseteq \mathbb{R}^n$ is bounded if there exists a $R > 0$, $A \subseteq \{x \in \mathbb{R}^n : ||x|| < R\}$.

Note that the complement of an unbounded set may not be bounded. Yet the complement of a bounded set is always unbounded.

Theorem 10.3 (Bolzano-Weierstrass)
A set being compact if and only if it is both closed and bounded.

We did not prove this in class. Generally we do not prove theorems in class.

[5] The typical definition is that a set $A \subseteq \mathbb{R}^n$ is compact if every open cover $\mathcal{O}$ (union of open sets such that $A \subseteq \mathcal{O}$) of A has finite subcover ($A \subseteq S \subsetneq \mathcal{O}$, where S has finitely many elements).

§11 Day 11: Limits (Sep 25, 2025)

"You get a delta, you get a delta, everybody gets a delta" - Q.

Definition 11.1 (Limit)**.** Let $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^m$ be a function. Let $a \in \mathbb{R}^n$ be a limit point of A, and let $b \in \mathbb{R}^m$. Define b to the limit of f at a provided that:

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } \forall x \in A, 0 < ||x - a|| < \delta \implies ||f(x) - b|| < \epsilon$$

If the above is true, then we write $\lim_{x \to a} f(x) = b$ or say $f(x) \to b$ as $x \to a$.

> **Remark 11.2.** In this course, if x left unquantified, then implicitly x is assumed to be an element of $\mathbb{R}^n$. Furthermore, we don't define limits at isolated points, because we want limits to be unique. Look at exercises 2.5.14, 2.5.15.

Having a be a limit point of A is important: without this condition, you could say that all points $b \in \mathbb{R}^m$ is the limit of f at any limit point $k \in \mathbb{R}^n$ that is not a limit point of A, since there would exist some δ-radius such that no points of A exist in the $B_\delta(k)$, thus the

Definition 11.3 (Isolated Point)**.** Have $A \subseteq \mathbb{R}^n$. A point $a \in \mathbb{R}^n$ is an isolated point of A if $a \in A$ and a is not a limit point of A.

This is equivalent to saying that a is an isolated point iff $a \in A \cap (\mathbb{R}^n \setminus A^*)$

Theorem 11.4 (Sequential Definition of Limits)
Let $A \subseteq \mathbb{R}^n$ be a set and let $f : A \to \mathbb{R}^m$ be a function. Let $a \in \mathbb{R}^n$ be a limit point of A and let $b \in \mathbb{R}^m$. Then $\lim_{x \to a} f(x) = b$ if and only if every sequence of points $\{x(k)\}_k$ in $A \setminus \{a\}$ with $x(k) \to a$, the sequence of points $\{f(x(k))\}_k$ in $\mathbb{R}^m$ converges to b, that is $f(x(k)) \to b$.

Proof was left as an exercise.

Theorem 11.5 (Component-wise Limits)
Let $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^m$ be a function. Let a be a limit point of A and $b = (b_1, \cdots, b_m) \in \mathbb{R}^m$. Let f_i be the component functions of f, such that $f = (f_1, \cdots, f_m)$, then $\lim_{x \to a} f(x) = b$ if and only if for all $i \in \{1, \cdots, m\}$, $\lim_{x \to a} f_i(x) = b$.

§12 Day 12: Continuity (Sep 29, 2025)

Definition 12.1 (Continuity)**.** Let $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^m$ be a function, let $a \in A$. f is continuous at a if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } \forall x \in A, ||x - a|| < \delta \implies ||f(x) - f(a)|| < \epsilon$$

With this definition, f is always continuous at isolated points in A, since there is a δ radius around said point that do not contain any other points in A. Yet we cannot write $\lim_{x \to a} f(x)$ here, since a would not be a limit point (from last class, we only define limits at limit points).

Theorem 12.2. If a is indeed a limit point, then $\lim_{x \to a} f(x) = f(a)$ is equivalent to our definition of continuity.

Theorem 12.3. Linear functions are continuous.

§13 Day 13: Path Connected Sets (Oct 06, 2025)

§14 Day 14: Global Extrema (Oct 06, 2025)

§15 Day 15: Derivatives of One Variable (Oct 06, 2025)

"Professors say: 'Never differentiate in public. You will only embarrass yourself."' -Q

Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}^m$ be a function. Let a be an interior point of A.

Definition 15.1 (Differential)**.** If f is differentiable at a, define the linear map $df_a : \mathbb{R} \to \mathbb{R}^m$ given by $df_a(h) = f'(a)h$ called the differential of f at a.

Remark 15.2. Vectors $v \in \mathbb{R}^m$ are identified with $m \times 1$ column vectors

$$\begin{pmatrix} v_1 \\ \vdots \\ v_m \end{pmatrix}$$

instead of $1 \times m$ row vectors, despite the fact that we write $v = (v_1, \cdots, v_m)$.

From first year calculus, the linear approximation of $f : \mathbb{R} \to \mathbb{R}$ at a point a is given by

$$g(x) = f(a) + f'(a)(x - a)$$

Substituting $x - a$ with h, we get that

$$f(a + h) \approx f(a) + \underbrace{hf'(a)}_{df_a(h)}$$

Rearranging, we get that $f(a + h) - f(a) - df_a(h) \approx 0$. In fact, we can strengthen this condition, such that

$$\lim_{h \to 0} \frac{f(a + h) - f(a) - L(h)}{h} = 0$$

where $L : \mathbb{R} \to \mathbb{R}^m$ is a linear transformation, such that $L(h) = f'(a)h = df_a(h)$.

Does this hold for general functions $\mathbb{R}^n \to \mathbb{R}^m$? Do we distinguish between linear transformations and matrices?

§16 Day 16: Partial Derivatives (Oct 07, 2025)

We now want to generalize 1 dimensional derivatives to n-dimensional derivatives. Unfortunately, we find that this is not the right generalization, for reasons you will see later.

Definition 16.1 (Partial Derivative)**.** The i-th partial derivative of $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^m$ at $a \in A^o$ is given by

$$\partial_j f(a) := \lim_{h \to 0} \frac{f(a + he_j) - f(a)}{h}$$

Equivalent notation for $\partial_j f$ are

$$\frac{\partial f}{\partial x_j}, \quad D_{e_j} f, \quad f_{x_j}, \quad D_j f, \partial_{x_j} f$$

Note that $\partial_j f(a)$ is essentially the limit of a real valued function, and is relatively straightforward to compute.

"Translating technical language into everyday language is a crucial skill that you will need later in your careers. It is best if you develop it during your studies."

§17 Day 17: Directional Derivatives (Oct 09, 2025)

Definition 17.1 (Directional Derivative)**.** This is a vector quantity. The directional derivative of f at a in the direction v is given by

$$D_v f(a) = \lim_{h \to 0} \frac{f(a + hv) - f(a)}{h}$$

It can be shown that $D_v f(a)$ exists if and only if $D_v f_i(a)$ exists for $i \in \{1, \cdots, m\}$.